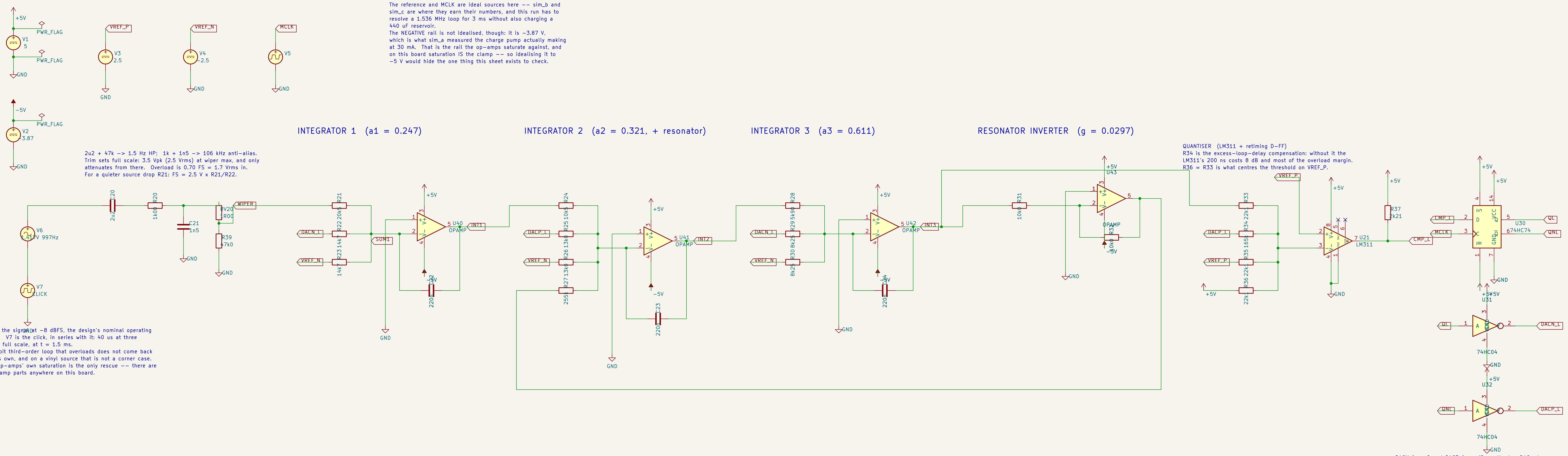
H One complete channel – 3rd-order CT delta-sigma, fs 1.536 MHz, OSR 32

The reference and MCLK are ideal sources here -- sim_b and sim_c are where they earn their numbers, and this run has to resolve a 1.536 MHz loop for 3 ms without also charging a 440 uF reservoir.
The NEGATIVE rail is not idealised, though: it is −3.87 V, which is what sim_a measured the charge pump actually making at 30 mA. That is the rail the op-amps saturate against, and on this board saturation IS the clamp -- so idealising it to −5 V would hide the one thing this sheet exists to check.



2u2 + 47k -> 1.5 Hz HP; 1k + 1n5 -> 106 kHz anti-alias.
Trim sets full scale: 3.5 Vpk (2.5 Vrms) at wiper max, and only attenuates from there. Overload is 0.70 FS = 1.7 Vrms in.
For a quieter source drop R21: FS = 2.5 V x R21/R22.

QUANTISER (LM311 + retiming D-FF)
R34 is the excess-loop-delay compensation: without it the LM311's 200 ns costs 8 dB and most of the overload margin.
R36 = R33 is what centres the threshold on VREF_P.

V6 is the signal at −8 dBFS, the design's nominal operating point. V7 is the click, in series with it: 40 us at three times full scale, at t = 1.5 ms.
A 1-bit third-order loop that overloads does not come back on its own, and on a vinyl source that is not a corner case.
The op-amps' own saturation is the only rescue -- there are no clamp parts anywhere on this board.

DACN from Q and DACP from /Q, so the two DAC edges are as close to simultaneous as the parts allow

Three things to read off this sheet, in order of how much they matter.

1. THE CLICK TEST. At 1.5 ms a 40 us transient at three times full scale hits the input. The loop must come back: the integrators must return to their normal 1.4–1.5 V swing and QL must go on toggling. A modulator that latches here is one that would need a power cycle after every click on the record.

2. WHERE THE TL074 ACTUALLY SATURATES. There are no clamp parts: the integrator state scales are chosen so the op-amps' own output saturation IS the clamp. The design assumed about +/−2.7 V on +/−5 V rails. If the real part swings further, the clamp lands somewhere the Python model never simulated -- so measure it, do not assume it.

3. INTEGRATOR SWINGS AT −8 dBFS: 1.43 / 1.47 / 1.45 V, from docs/design-notes.md 3. Bigger than that and the overload margin has gone somewhere; smaller and the quantisation noise is being spent for nothing.

There is deliberately NO SNR figure here. 68 dB over a 20 kHz band at OSR 32 needs about a second of transient at nanosecond steps; sim/verify.py computes it properly in minutes and this sheet would take hours to say the same.

.tran 10n 3m uic

<- back to the board schematic   vinyl_adc.kicad_sch

Sheet: /	
File: sim_h_loop.kicad_sch	
Title: H – one complete modulator channel, with the click test	
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